EXP NO:14 Experiment to understand feature scaling.

AIM:
To understand the importance of feature scaling.

PROGRAM:

python

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler, StandardScaler

# Creating dataset with features on different scales
data = {
    'Age': [18, 22, 25, 30, 35, 40, 45],
    'Salary': [20000, 35000, 50000, 80000, 120000, 150000, 200000]
}
df = pd.DataFrame(data)
df

# Statistical description before scaling
df.describe()

# Min-Max Scaling
minmax = MinMaxScaler()
minmax_scaled = pd.DataFrame(
    minmax.fit_transform(df),
    columns=df.columns
)
minmax_scaled

# Statistical description after Min-Max Scaling
minmax_scaled.describe()

# Standardization (Z-score scaling)
standard = StandardScaler()
standard_scaled = pd.DataFrame(
    standard.fit_transform(df),
    columns=df.columns
)
standard_scaled

# Statistical description after Standardization
standard_scaled.describe()

# Compare original and scaled data
fig, axes = plt.subplots(1, 3, figsize=(15, 4))

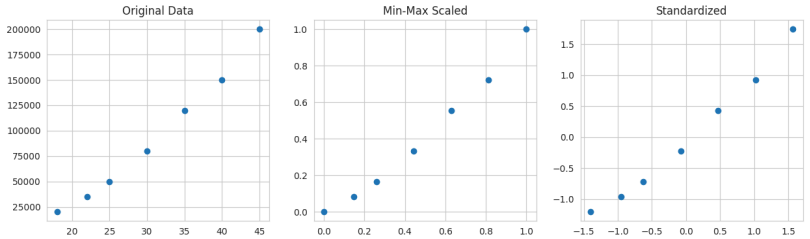
axes[0].scatter(df['Age'], df['Salary'])
axes[0].set_title("Original Data")

axes[1].scatter(minmax_scaled['Age'], minmax_scaled['Salary'])
axes[1].set_title("Min-Max Scaled")

axes[2].scatter(standard_scaled['Age'], standard_scaled['Salary'])
axes[2].set_title("Standardized")

plt.show()
```

OUTPUT:



RESULT:
Thus the program to understand the importance of feature scaling using Min-Max scaling and Standardization was executed successfully.